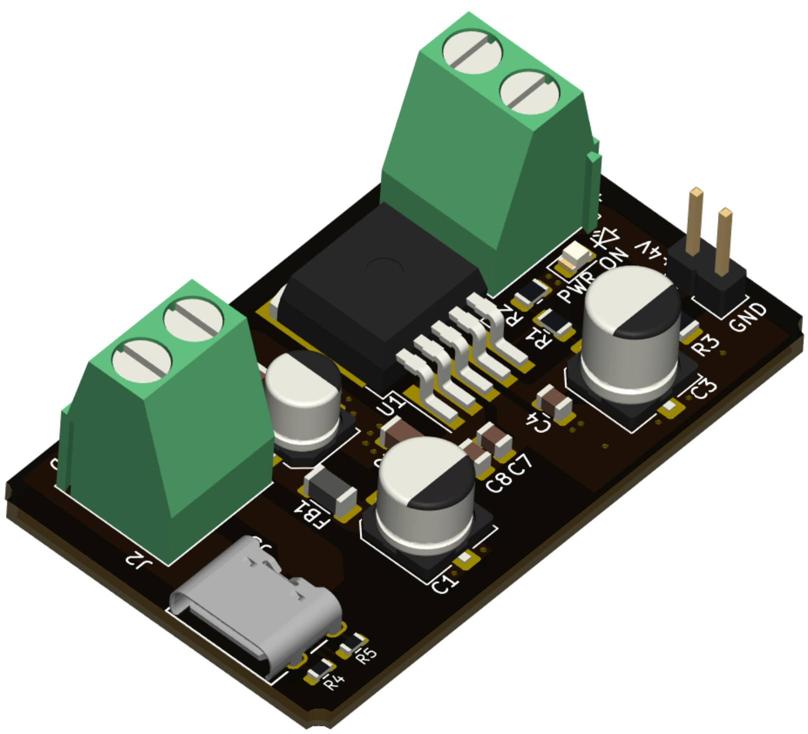
SIM800L Power Supply Evaluation Board

Hardware Development & Evaluation Kit

Revision 1.0



1. DESCRIPTION

The SIM800L Power Supply Evaluation Board is a hardware development platform specifically designed to provide a stable and low-noise power source for GSM/GPRS modules based on the SIM800L chipset.

The board is built around the high-current low-dropout regulator MIC29302WU and has been optimized to satisfy the dynamic current requirements specified by the SIM800L hardware design guidelines, including transmission bursts up to 2 A peak current.

The design incorporates:

- MIC29302WU High-Current LDO Regulator
- Input EMI Pi Filter
- Panasonic Solid Polymer Capacitors
- Würth Elektronik Ferrite Bead
- Molex USB-C Power Input Connector
- Phoenix THT Power Connectors
- Blue Power Status LED
- Customizable RF EMI Network

1.1 Application Compatibility

Although specifically optimized for the SIM800L GSM/GPRS module, this evaluation board may also be used with other GSM/GPRS cellular modules requiring a regulated supply voltage in the 4.0 V to 4.2 V range and capable of handling transmission current peaks up to 2 A.

Typical compatible modules include, but are not limited to:

- SIM800L
- SIM800C
- SIM800H
- SIM808

- SIM868
- A6 GSM/GPRS Module
- A7 GSM/GPRS/GPS Module
- M66 GSM/GPRS Module
- M95 GSM/GPRS Module

Before connecting any third-party module, the user shall verify that the operating voltage range, peak current requirements, startup characteristics, and transient load conditions are compatible with the electrical specifications of this evaluation board.

The suitability of this board for modules from manufacturers other than SIMCom shall be validated by the user during system integration and testing.

⚠ EVALUATION HARDWARE NOTICE

This hardware platform is provided solely as an engineering evaluation and development tool.

It is intended for use by engineers, technicians, researchers, and experienced electronics developers for laboratory testing, GSM modem evaluation, proof-of-concept validation, embedded system development, industrial prototyping, and educational activities.

The board is not designed, tested, certified, or approved as a finished consumer product and shall not be used directly in commercial, safety-critical, automotive, medical, life-support, or other regulated applications without appropriate system-level validation, certification, and regulatory compliance assessment performed by the integrator.

2. TECHNICAL SPECIFICATIONS

Parameter	Value
Input Voltage Range	5 V to 12 V DC
Recommended Input Voltage	5 V to 9 V DC
Output Voltage	4.1 V DC Typical
Output Voltage Range	4.0 V – 4.2 V
Maximum Continuous Output Current	2 A*
Peak Output Current	2 A GSM Burst
Regulator Type	MIC29302WU LDO
Dropout Voltage	Refer to MIC29302WU Datasheet
Input Filter	Pi EMI Network
Polarized Capacitors	Panasonic Solid Polymer
Status Indicator	Blue Power LED
PCB Technology	FR4, 2-Layer
Operating Temperature	0°C to +70°C

*Thermal limitations depend on ambient conditions and available PCB copper area.

3. PINOUT AND MECHANICAL CONNECTIONS

3.1 Input Power Phoenix Connector:

Primary power input.

Pin	Name	Description
1	+5V	Main DC Input Voltage
2	GND	Common Ground

Electrical Requirements

Parameter	Value
Minimum VIN	5 V
Recommended VIN	**5 V (strongly recommended)** - 9 V
Maximum VIN	12 V not recommended

Although the **MIC29302WU** supports higher voltages, operation above **12 V is not recommended** due to increased power dissipation and regulator temperature rise.

3.2 Regulated Output Phoenix Connector

Dedicated output for SIM800L power supply.

Pin	Name	Description
1	+4V	Regulated Output
2	GND	Power Ground

Output Characteristics

Parameter	Value
Minimum Voltage	4.0 V
Nominal Voltage	**4.1 V**
Maximum Voltage	4.2 V
Intended Load	SIM800L GSM Module

The output voltage is established through the MIC29302WU feedback resistor network and is optimized for SIM800L operation.

3.3 USB-C Power Interface

The USB-C connector provides an alternative power input source.

Supported Signals

Pin Group	Function
VBUS	+5V Input
GND	Ground
CC1/CC2	USB-C Power Detection
D+	Not Connected
D−	Not Connected

The USB-C connector is intended exclusively for power delivery.

No USB communication functionality is implemented.

4. INPUT EMI FILTERING NETWORK

4.1 Description

The evaluation board incorporates a fixed input EMI suppression stage based on a Pi-filter topology, consisting of a ferrite bead and low-impedance capacitors specifically selected to attenuate conducted noise and improve power integrity.

To further optimize RF immunity in demanding installations, two optional capacitor positions (C_EMI1 and C_EMI2) are provided at the input of the MIC29302WU regulator. These components allow the user to tailor the high-frequency response of the input filter to specific GSM operating bands and installation conditions.

Depending on antenna placement, enclosure design, PCB integration, and EMC requirements, one or both capacitors may be populated to enhance suppression of RF energy coupled onto the power distribution network.

4.1 Purpose

The SIM800L operates in GSM frequency bands around:

- GSM900: 880 MHz – 960 MHz
- DCS1800: 1710 MHz – 1880 MHz

During RF transmission bursts, the antenna radiates high-energy electromagnetic fields that may couple into nearby PCB traces and power rails.

If these RF signals reach the MIC29302WU regulator input, the internal feedback circuitry may unintentionally demodulate the RF carrier.

The resulting low-frequency envelope can appear as audible or measurable disturbances around 217 Hz, corresponding to the GSM TDMA burst repetition frequency.

This phenomenon may degrade system stability, increase output ripple, and interfere with nearby circuitry.

To improve RF immunity, two **optional SMD capacitor** positions have been provided directly at the regulator input.

Both capacitors are connected between:

MIC29302WU Pin 2 (VIN) and MIC29302WU Pin 3 (GND)

4.2 C_EMI1 C7 – GSM900 Suppression Capacitor

Recommended Component

Parameter	**Value**
Type	MLCC
Dielectric	NP0 / C0G
Package	**0603 (Recommended)** or 0805
Typical Value	33 pF – 39 pF
Target Frequency	~900 MHz

Function

Provides a low impedance path to ground for RF energy present in the GSM900 band before it can reach the regulator input stage.

4.3 C_EMI2 C8 – DCS1800 Suppression Capacitor

Recommended Component

Parameter	Value
Type	RF Grade MLCC
Dielectric	NP0 / C0G
Package	**0603 (Recommended)** or 0805
Typical Value	8.2 pF – 10 pF
Target Frequency	~1.8 GHz

Function

Provides attenuation of RF energy present in the DCS1800 band.

When used together with C_EMI1, broadband RF suppression is obtained across the primary GSM transmission frequencies.

4.4 Installation Notes

Recommended placement order:

1. Install C_EMI1 first.
2. Perform RF emission testing.
3. Install C_EMI2 if additional immunity is required.

Use only NP0/C0G dielectric capacitors to ensure predictable impedance and stable self-resonant frequency characteristics.

5. TESTING AND VALIDATION PROCEDURE

5.1 Initial Power-Up

Before connecting the SIM800L:

1. Verify assembly quality.
2. Inspect solder joints.
3. Confirm capacitor polarity.
4. Verify absence of short circuits.

5.2 No-Load Verification

Connect a laboratory power supply to the input connector.

Recommended initial settings:

Parameter	Value
V_{IN}	**+5V (Strongly recommended)** or 9 V
Current Limit	200 mA

Power the board and verify that the blue LED illuminates correctly.

5.3 Static Output Measurement

Using a calibrated digital multimeter, measure between: +4V and GND

Expected result:

Parameter	Value
Minimum	3.9 V
Typical	**4.1 V**
Maximum	4.2 V

Any deviation outside this range should be investigated before connecting the GSM module.

5.4 Dynamic Load Analysis

Connect the SIM800L module.

Configure the oscilloscope:

Setting	Value
Coupling	AC
Bandwidth	Full
Probe	10X
Trigger	Falling Edge

Monitor +4V VOUT during GSM transmission bursts.

Acceptance Criteria

Parameter	Limit
Maximum Voltage Drop	<300 mV
Minimum Transient Voltage	>3.4 V
Oscillation	None
Recovery Time	Immediate

If the voltage falls below 3.4 V, SIM800L hardware shutdown may occur.

6. THERMAL CONSIDERATIONS

Power dissipated by the regulator is approximately: **$P = (V_{IN} - V_{OUT}) \times I_{OUT}$**

Examples:

V_{IN}	I_{OUT}	Dissipation
5 V	**1 A**	**0.9 W**
9 V	1 A	4.9 W
12 V	1 A	7.9 W

For continuous operation above 1 A, input voltages near 5 V are strongly recommended to minimize thermal stress.

7. PRODUCT DISCLAIMER

This electronic module is a bare development kit intended exclusively for research, development, prototyping and evaluation activities in laboratory environments by qualified personnel or experienced hobbyists. It is not a finished product intended for general domestic or commercial use.

The module has not been subjected to formal laboratory testing for CE, EMC or regulatory certification.

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The company also specializes in advanced embedded firmware development for real-time systems, including bare-metal architectures, finite state machine (FSM) frameworks, hardware abstraction layers (HAL), low-level firmware, industrial communication protocols, and real-time control applications.
Core competencies include mixed-signal electronics, power conversion, embedded Linux, ARM Cortex-M microcontrollers, hardware/software co-design, rapid prototyping, system validation, and custom engineering solutions for industrial and IoT applications.

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Repository: https://github.com/SCD-Labs/PWR-SIMM800L-IND-v1.0

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